

The Oxidative Tricarboxylic Acid Cycle: A Taxon-Neutral Review of Boundaries, Mechanism, Variation, and Origin

1. Executive Summary

The oxidative tricarboxylic acid (TCA, citric acid, or Krebs) cycle is a closed, eight-reaction ring that oxidizes a two-carbon acetyl unit — delivered as acetyl-CoA — to two molecules of CO₂ while regenerating its four-carbon acceptor, oxaloacetate. Along the way it captures reducing equivalents (three NAD(P)H-equivalents and one membrane-quinol) and performs a single substrate-level phosphorylation. The cycle sits at the metabolic crossroads of catabolism (feeding electrons to the respiratory chain) and biosynthesis (exporting citrate, 2-oxoglutarate, succinyl-CoA, and oxaloacetate as biosynthetic precursors). Its reaction *topology* — the order of chemical transformations — is remarkably conserved and, on both chemical and thermodynamic grounds, largely forced: condensation must precede isomerization, which must precede the two oxidative decarboxylations, which must precede the C₄ dicarboxylate oxidations that regenerate oxaloacetate.

Despite this conserved chemistry, the *molecular solutions* that implement each step are strikingly lineage-variable. Almost every node of the cycle has documented alternative enzymes or bypasses: fumarate hydration is catalyzed by two structurally unrelated protein classes (a non-orthologous gene displacement); malate oxidation can proceed through soluble NAD-dependent malate dehydrogenase or through a membrane-bound, quinone-coupled malate:quinone oxidoreductase; succinyl-CoA synthetase exists in ATP- and GTP-specific isoforms distinguished by "gatekeeper" residues; and cyanobacteria, which lack the 2-oxoglutarate dehydrogenase complex, close the cycle through a 2-oxoglutarate decarboxylase / succinic-semialdehyde dehydrogenase bypass supplemented by a GABA shunt. Several cycle enzymes moonlight: cytosolic aconitase is iron regulatory protein-1 (IRP1), a post-transcriptional iron sensor, and class-I fumarase participates in the DNA damage response.

The cycle is also deeply ancient and medically central. Its reverse (reductive) form is one of the oldest CO₂-fixation pathways and can be partially driven by metal ions in the absence of enzymes, supporting a prebiotic, proto-metabolic origin. In humans, gain-of-function

mutations in isocitrate dehydrogenase (IDH1/2) and loss-of-function mutations in succinate dehydrogenase (SDH) and fumarate hydratase (FH) are bona fide cancer drivers that act through oncometabolites (D-2-hydroxyglutarate, succinate, fumarate) that inhibit 2-oxoglutarate-dependent dioxygenases and remodel the epigenome. The most robustly supported claims concern the core chemistry, the ordered/irreversible steps, and oncometabolite biology; the principal open questions concern the physiological reality of a physically organized "metabolon," the detailed catalytic mechanics of the swinging-domain enzymes, and how far lineage-specific regulatory logic can be generalized.

2. Definition and Biological Boundaries

2.1 What is included

The module treated here is the **oxidative** TCA cycle: acetyl-CoA entry through **citrate synthase**; citrate → isocitrate isomerization by **aconitase**; oxidative decarboxylation of isocitrate to 2-oxoglutarate by **isocitrate dehydrogenase**; oxidative decarboxylation of 2-oxoglutarate to succinyl-CoA by the **2-oxoglutarate dehydrogenase complex** (E1/E2/E3); substrate-level phosphorylation by **succinyl-CoA synthetase**; succinate oxidation to fumarate by **succinate dehydrogenase / respiratory complex II**; fumarate hydration by **fumarate hydratase**; and malate oxidation to oxaloacetate by **malate dehydrogenase**. Two boundary elements are included by design: the bacterial **malate:quinone oxidoreductase (MQO)** alternative at the malate/oxaloacetate node, and **pyruvate carboxylase** as an anaplerotic input to oxaloacetate.

2.2 What should be treated separately

Several neighboring processes are frequently conflated with the cycle but are mechanistically and definitionally distinct:

- **Pyruvate dehydrogenase complex (PDC).** The PDC generates acetyl-CoA from pyruvate and is the cycle's principal feed, but it is not a cycle reaction. It is structurally homologous to the 2-oxoglutarate dehydrogenase complex (shared E3, lipoyl chemistry) which invites confusion.
- **The respiratory electron transport chain.** Only succinate dehydrogenase (Complex II) is physically shared between the cycle and oxidative phosphorylation. The remaining oxidative payoff (re-oxidation of NADH, proton pumping, ATP synthase) is downstream and separable.

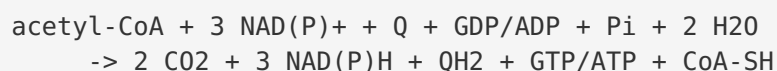
- **The glyoxylate shunt.** Isocitrate lyase + malate synthase bypass the two oxidative decarboxylation steps, conserving carbon for gluconeogenesis on C2 substrates. This is a regulated *bypass*, not part of the oxidative cycle proper.
- **Reductive (reverse) TCA cycle.** Running the same reaction set backward for CO₂ fixation is a distinct physiological mode requiring different thermodynamic driving and, at some steps, different enzymes.
- **Anaplerosis / cataplerosis.** Pyruvate carboxylase and glutaminolysis replenish intermediates; citrate export to cytosolic acetyl-CoA (via ATP-citrate lyase) drains them. These flank the cycle rather than constitute it.

2.3 Competing definitions

Literature usage varies on whether the cofactor specificity of isocitrate dehydrogenase (NAD⁺ vs NADP⁺) and the nucleotide specificity of succinyl-CoA synthetase (ATP vs GTP) should be considered defining. Because these differ between and within lineages, the taxon-neutral definition treats the *transformation* (oxidative decarboxylation; substrate-level phosphorylation) as canonical while allowing the cofactor identity to vary. Similarly, whether the "incomplete" cyanobacterial cycle should be called a TCA cycle at all was long contested until the discovery of its bypass reactions (Section 5.2).

3. Mechanistic Overview

The best-supported model is a strictly ordered ring in which chemistry and thermodynamics enforce step sequence. The overall transformation per turn:



Step-by-step:

[1] Citrate synthase	acetyl-CoA + OAA + H ₂ O → citrate + CoA	(irreversible; ordered)
[2] Aconitase	citrate ↔ isocitrate	(Fe-S; near-equilibrium)
[3] Isocitrate DH	isocitrate → 2-oxoglutarate + CO ₂	(first oxidative decarboxylation)
[4] 2-oxoglutarate DH cx	2-OG + CoA → succinyl-CoA + CO ₂	(E1/E2/E3; second oxidative decarboxylation)
[5] Succinyl-CoA synthetase	succinyl-CoA + Pi + GDP/ADP ↔ succinate	(only substrate-level phosphorylation)
[6] Succinate DH / CII	succinate + Q → fumarate + QH ₂	(membrane-linked; feeds into electron transport chain)
[7] Fumarate hydratase	fumarate + H ₂ O ↔ L-malate	(stereospecific hydration)
[8] Malate DH	L-malate + NAD ⁺ ↔ OAA + NADH	(regenerates acceptor for succinate DH)
Anaplerosis: pyruvate + HCO ₃ ⁻ + ATP → OAA (pyruvate carboxylase)		

Obligatory steps are the two oxidative decarboxylations (steps 3–4), which are the carbon-removing, CO₂-releasing heart of the oxidative cycle, and the acetyl-condensation entry (step 1). **Conditional / variant steps** include the substrate-level phosphorylation (nucleotide identity varies; bypassed entirely in cyanobacteria) and malate oxidation (NAD- vs quinone-coupled). **Accessory** processes are anaplerotic input (pyruvate carboxylase), the glyoxylate bypass, and citrate cataplerosis.

Three steps are effectively irreversible under physiological conditions and act as flux-control and regulatory points: citrate synthase (highly exergonic owing to citryl-CoA hydrolysis), isocitrate dehydrogenase, and the 2-oxoglutarate dehydrogenase complex. The remaining reactions run near equilibrium, allowing the same enzymes, in principle, to operate in reverse when thermodynamics permit.

4. Major Molecular Players and Active Assemblies

4.1 Citrate synthase — ordered induced-fit entry

Citrate synthase enforces ordered sequential binding: oxaloacetate (OAA) binds first and triggers a large open→closed domain motion that completes the acetyl-CoA site. A conserved active-site histidine (His320 in pig heart; His264 in *E. coli*) polarizes the OAA carbonyl to activate the aldol-type Claisen condensation, which proceeds through a citryl-CoA intermediate that is then hydrolyzed — making the overall reaction strongly exergonic and effectively irreversible ([PMID: 7577912](#); [PMID: 8010958](#)). Allosteric control is lineage-specific: Gram-negative bacterial citrate synthase is a hexamer allosterically inhibited by NADH, whereas eukaryotic and Gram-positive enzymes are non-allosteric dimers ([PMID: 23954305](#)). This is finding **F011**.

4.2 Aconitase — a moonlighting Fe-S switch

Aconitase uses a cubane [4Fe-4S] cluster for the citrate↔isocitrate isomerization. In the cytosol, the enzyme is identical to **iron regulatory protein-1 (IRP1)**: its two activities are mutually exclusive and toggled by the cluster. The cluster-bearing form is an active aconitase; the apo (cluster-less) form binds iron-responsive elements (IREs) in the UTRs of ferritin and transferrin-receptor mRNAs, controlling their translation and stability. Iron depletion, nitric oxide, and oxidative stress (H₂O₂, quinone redox cycling) disassemble the cluster and switch function post-translationally and reversibly ([PMID: 8682202](#); [PMID: 10037708](#)). This is finding **F004** and a paradigm of enzyme moonlighting.

4.3 Isocitrate dehydrogenase — a regulated flux valve

The first oxidative decarboxylation yields 2-oxoglutarate and CO₂. In enterobacteria, the bifunctional **isocitrate dehydrogenase kinase/phosphatase (AceK)** phosphorylates a substrate-binding serine of IDH to inactivate it, diverting isocitrate into the glyoxylate shunt — historically the *first identified prokaryotic protein phosphorylation* ([PMID: 22889914](#); [PMID: 30030382](#)). IDH homodimerization combined with AceK bifunctionality confers robustness to the flux switch ([PMID: 23192354](#)). Regulation is lineage-specific: *E. coli* IDH is also lysine-acetylated ([PMID: 29733852](#)), and eukaryotic IDHs share no sequence similarity with the *E. coli* enzyme and use different controls ([PMID: 33290880](#)). This is finding **F009**.

4.4 The 2-oxoglutarate dehydrogenase complex — a giant multienzyme machine

The second oxidative decarboxylation is carried out by a large E1–E2–E3 assembly: E1 (2-oxoglutarate dehydrogenase) decarboxylates the substrate; E2 (dihydrolipoyllysine-residue succinyltransferase) transfers the succinyl group to CoA using a swinging lipoyl-lysine arm; and E3 (dihydrolipoyl dehydrogenase) re-oxidizes the lipoamide, reducing NAD⁺. E3 and the lipoyl chemistry are shared with the pyruvate dehydrogenase complex ([PMID: 8720178](#)). The complex is a highly regulated redox sensor sensitive to reactive oxygen species and to the oncometabolite D-2-hydroxyglutarate, which allosterically inhibits it ([PMID: 35530286](#); [PMID: 42327104](#)).

4.5 Succinyl-CoA synthetase — the cycle's only substrate-level phosphorylation

Succinyl-CoA synthetase (SCS) catalyzes the sole substrate-level phosphorylation of the cycle through a phosphohistidine intermediate that physically shuttles ~29 Å between two active sites; succinyl-phosphate is a genuine catalytic intermediate captured at 1.58 Å resolution (PMID: 33645539; PMID: 36189720; PMID: 27487822). Nucleotide specificity (ADP/ATP vs GDP/GTP) is governed by solvent-exposed **gatekeeper residues**; engineering these residues reverses specificity (PMID: 27478903). Mammals express distinct ATP- and GTP-specific β -subunit isoforms, linking SCS to a mitochondrial GTP signaling cycle that couples anaplerosis to insulin secretion (PMID: 31315053). The enzyme is also the mechanistic prototype for ATP-citrate lyase. This is finding **F003**.

4.6 Succinate dehydrogenase / Complex II — the membrane-linked node

SDH couples succinate oxidation in the cycle to quinone reduction in the respiratory chain, making it the *only membrane-embedded cycle enzyme* and the sole enzyme shared with oxidative phosphorylation. Its structure varies across lineages: the mammalian "type C" carries a heme b; yeast Complex II has lost the canonical heme; and mycobacterial Sdh1 is a distinct "type F" with a membrane-embedded Rieske-type FeS cluster (PMID: 33876763; PMID: 41606097). Beyond respiration, Complex II activity participates in metabolic control, inflammation, and cell-fate signaling via succinate, and it assembles into megacomplexes in cell-type-specific ways (PMID: 37119852; PMID: 34563205). This is finding **F006**.

4.7 Fumarate hydratase — two unrelated protein classes

Fumarate hydration is a striking case of **non-orthologous gene displacement**: two structurally unrelated enzyme classes catalyze the same reaction. Class I fumarases (FumA/FumB) are oxygen-labile homodimers carrying a [4Fe-4S] cluster, related to aconitase-type iron-dependent hydro-lyases; Class II fumarases (FumC and the porcine/yeast/*B. subtilis* enzymes) are thermostable tetramers ~60% identical to one another and homologous to aspartase and argininosuccinase, with no significant sequence similarity to Class I (PMID: 3282546; PMID: 1329945; PMID: 8473853). Eukaryotic Class II fumarase is dual-targeted to mitochondria and cytosol and moonlights in the DNA damage response (PMID: 34083440). This is finding **F005**.

4.8 Malate oxidation — soluble vs quinone-coupled

The regeneration of oxaloacetate can proceed by two non-homologous enzymes. Soluble NAD-dependent malate dehydrogenase (EC 1.1.1.37) is reversible and thermodynamically unfavorable in the forward direction. The membrane-associated **malate:quinone oxidoreductase (MQO; EC 1.1.99.16)** is an FAD flavoprotein that donates electrons directly to quinone, coupling the step to the respiratory chain and rendering it effectively irreversible. In *Corynebacterium glutamicum* MQO is the principal malate oxidase; in *E. coli* both are expressed (mqo under ArcA-ArcB control), and mdh deletion severely impairs growth ([PMID: 11092847](#); [PMID: 9660197](#)). This is finding **F007**.

4.9 Pyruvate carboxylase — biotin-dependent anaplerosis

Pyruvate carboxylase (PC) carboxylates pyruvate to oxaloacetate in a bicarbonate- and ATP-dependent, biotin-mediated reaction, replenishing cycle intermediates. It is allosterically activated by acetyl-CoA (signaling a need to make OAA) and inhibited by L-aspartate (signaling sufficient OAA-derived output); the acetyl-CoA site maps to the biotin-carboxylase dimer interface ([PMID: 37603581](#); [PMID: 39725256](#)). PC and glutamine→2-oxoglutarate are the two dominant anaplerotic routes, and PC-dependent anaplerosis is context-specific in cancer, being upregulated in highly invasive breast cancer cells and lung metastases ([PMID: 27732858](#); [PMID: 27890529](#)). This is finding **F008**.

4.10 Summary table

Step	Enzyme	Transformation	Key feature / variant
1	Citrate synthase	acetyl-CoA + OAA → citrate	Ordered induced fit; citryl-CoA intermediate; hexamer (bacterial, NADH-inhibited) vs dimer
2	Aconitase	citrate ↔ isocitrate	[4Fe-4S]; cytosolic form = IRP1 (moonlighting)
3	Isocitrate dehydrogenase	isocitrate → 2-OG + CO ₂	NAD ⁺ vs NADP ⁺ ; AceK phospho-switch to glyoxylate bypass
4	2-OG dehydrogenase complex	2-OG → succinyl-CoA + CO ₂	E1/E2/E3 machine; redox sensor; inhibited by D2-HG
5	Succinyl-CoA synthetase	succinyl-CoA → succinate	Only substrate-level phosphorylation; ATP vs GTP gatekeepers
6	Succinate dehydrogenase / CII	succinate → fumarate	Only membrane enzyme; type C/F variants; signaling roles
7	Fumarate hydratase	fumarate ↔ malate	Two unrelated classes (I Fe-S; II aspartase-like)
8	Malate dehydrogenase / MQO	malate → OAA	Soluble NAD vs membrane quinone-coupled
+	Pyruvate carboxylase	pyruvate → OAA	Biotin-dependent anaplerosis; acetyl-CoA/aspartate control

5. Evolutionary and Cell-Biological Variation

5.1 Deep origin

The **reductive (reverse) TCA cycle** is among the most ancient autotrophic CO₂-fixation metabolisms and operates in deeply branching chemolithoautotrophs; in at least one thermophile a single enzyme set runs the cycle reversibly ([PMID: 29420286](#)). Strikingly, 6 of 11 rTCA reactions can be promoted non-enzymatically by metal ions such as Zn, Fe, and Cr, and

thermodynamic network models converge on a thioester- and redox-driven rTCA variant as the core of an early proto-metabolism ([PMID: 28970480](#); [PMID: 31712697](#)). This is finding **F002**, and it implies that the cycle's reaction *topology* predates its enzymes — the enzymes were later, lineage-specific recruitments onto a pre-existing chemical scaffold.

5.2 Non-orthologous displacements and alternative routes

The recurring theme is that the *chemistry* is conserved but the *catalyst* is not. The two fumarase classes (F005) and the NAD-MDH-vs-MQO malate node (F007) are the clearest non-orthologous displacements. The ATP/GTP SCS isoforms (F003) and the NAD/NADP IDH variants illustrate cofactor plasticity on a conserved scaffold. The most dramatic rewiring is in **cyanobacteria**, which lack the 2-oxoglutarate dehydrogenase complex and instead convert 2-oxoglutarate to succinate via 2-oxoglutarate decarboxylase plus succinic-semialdehyde dehydrogenase — functionally replacing *both* OGDH and SCS — supplemented by a parallel GABA shunt that carries substantial flux in *Synechocystis* ([PMID: 22174252](#); [PMID: 24989231](#)). This is finding **F010**. These bypass genes are present in most cyanobacterial genomes (except *Prochlorococcus*/marine *Synechococcus*), and their discovery revised the long-held view that the cyanobacterial cycle was simply "incomplete."

5.3 Compartmental and cell-type variation

In eukaryotes the oxidative cycle is mitochondrial, but several players have cytosolic counterparts (cytosolic aconitase/IRP1; dual-targeted Class II fumarase; cytosolic IDH1). Complex II composition (heme content, megacomplex assembly) and SDH subunit expression are cell-type-specific and under signaling control — for example, CAMKK2 regulates SDH expression, post-translational modification, and megacomplex assembly differently in kidney-derived versus hepatoma cells ([PMID: 34563205](#)). PC-dependent anaplerosis is organ- and tumor-context-specific ([PMID: 27732858](#)). Thus, while the ring is universal, its regulation, localization, and quantitative wiring are tuned to physiological state.

6. Constraints, Dependencies, and Failure Modes

6.1 Ordering constraints

Step order is enforced by chemistry and thermodynamics, not by convention. Condensation (citrate synthase) must occur before isomerization, because isocitrate cannot be produced without citrate; aconitase-catalyzed isomerization must precede oxidative decarboxylation,

because the tertiary hydroxyl of citrate cannot be oxidized whereas the secondary hydroxyl of isocitrate can — this is precisely why the cell "wastes" a step isomerizing citrate to isocitrate. The two oxidative decarboxylations must precede the C4 oxidations. The near-irreversibility of citrate synthase (citryl-CoA hydrolysis), IDH, and OGDH biases net flux forward and makes these the committed control points.

6.2 Mutually exclusive and compartment-specific events

- Aconitase catalysis and IRE-binding are mutually exclusive functions of one protein, gated by the [4Fe-4S] cluster (F004).
- Full-cycle flux and the glyoxylate bypass are mutually exclusive at the isocitrate node, switched by IDH phosphorylation (F009).
- Complex II is intrinsically membrane-bound; the succinate→fumarate step cannot occur in a purely soluble context, tying it to the respiratory chain (F006).
- Quinone-coupled malate oxidation (MQO) and NAD-linked oxidation (MDH) draw on different electron acceptors and therefore have different reversibility and thermodynamic consequences (F007).

6.3 Failure modes and disease

Cycle-enzyme dysfunction produces oncometabolites. Gain-of-function IDH1/2 mutations generate D-2-hydroxyglutarate; loss-of-function SDH and FH mutations accumulate succinate and fumarate. These metabolites competitively inhibit 2-oxoglutarate-dependent dioxygenases (PHD2, TET, KDM demethylases), stabilizing HIF-1 α and remodeling DNA/histone methylation; D2-HG additionally allosterically inhibits the 2-oxoglutarate dehydrogenase complex (PMID: 31085323; PMID: 21301796; PMID: 42327104). This is finding **F001**. High fumarate/succinate also cause apoptotic cytotoxicity and global DNA-methylation changes in vitro (PMID: 28104508), and structurally similar metabolites (e.g., succinylacetone) can mimic these effects on HIF signaling (PMID: 37906252). These findings redefined TCA enzymes as tumor suppressors/oncogenes rather than mere housekeeping catalysts.

7. Controversies and Open Questions

1. **The "metabolon."** Whether cycle enzymes physically associate to channel substrates in vivo remains unresolved. Supporting evidence is largely from in vitro immobilization and FRET biomimics (PMID: 23848576) rather than direct in-cell structural proof. The functional importance of channeling for flux is uncertain.

2. **Swinging-domain catalytic mechanics.** The precise choreography of the SCS phosphohistidine shuttle and its relationship to the ATP-citrate lyase mechanism is still being refined from crystallographic snapshots (PMID: 36189720; PMID: 33645539). How the two active sites are kinetically coordinated in vivo is not fully settled.
3. **Generalizability of regulatory logic.** The AceK phospho-switch and NADH-inhibited hexameric citrate synthase are enterobacterial/Gram-negative features; eukaryotic enzymes share little sequence identity and use different controls (PMID: 33290880; PMID: 29733852). Extrapolating one organism's regulation to "all biology" is a recurring literature hazard.
4. **Physiological role of moonlighting.** The quantitative contribution of aconitase/IRP1 and Class-I fumarase moonlighting to iron homeostasis and DNA repair in vivo, versus their canonical metabolic roles, needs cleaner separation.
5. **Non-canonical Complex II biology.** Whether succinate-driven and megacomplex-associated signaling by Complex II is a widespread regulatory mode or a context-restricted phenomenon is an active question (PMID: 37119852).
6. **Prebiotic plausibility.** Non-enzymatic promotion of 6/11 rTCA reactions is compelling but incomplete; whether a *self-sustaining* metal-catalyzed cycle existed remains debated (PMID: 28970480).

8. Limitations and Knowledge Gaps

This review synthesizes findings from targeted literature (60 papers) rather than a single primary dataset; it is a knowledge-integration exercise, and the strength of each claim reflects the strength of its underlying primary literature. Several specific limitations should be flagged:

- Much mechanistic detail derives from a small number of model organisms (*E. coli*, pig heart, yeast, cyanobacteria). Cofactor and regulatory claims are frequently organism-specific and were treated as such.
- Oncometabolite mechanisms are strongly supported for IDH/SDH/FH but the relative contributions of HIF stabilization versus direct epigenetic remodeling remain quantitatively uncertain.
- The evolutionary origin arguments rest on comparative genomics and abiotic chemistry experiments that are suggestive rather than definitive; alternative origin scenarios exist.
- Quantitative flux data (isotope tracing across states/tissues) were outside the scope of the literature reviewed and would sharpen the variation claims.

9. Proposed Follow-up Actions

1. **Resolve the metabolon question in vivo** using proximity-labeling (BioID/APEX) or cryo-electron tomography in intact mitochondria across metabolic states, paired with isotope-channeling assays.
2. **Systematic gatekeeper/cofactor swaps** across SCS and IDH orthologs to map how frequently nucleotide/cofactor specificity has switched during evolution, and whether it correlates with lifestyle.
3. **Comparative structural survey of Complex II classes** (type C vs F vs heme-less yeast) to define the minimal quinone-coupling architecture and the origin of the type-F Rieske cluster.
4. **Flux partitioning at variant nodes** (MQO vs MDH; canonical OGDH vs cyanobacterial bypass vs GABA shunt) via ^{13}C metabolic flux analysis under defined growth conditions.
5. **Dissect moonlighting cleanly** with separation-of-function mutants that abolish aconitase catalysis while preserving IRE binding (and vice versa), and analogous Class-I fumarase DDR-vs-catalysis separations, to quantify each role in vivo.
6. **Oncometabolite target mapping** using chemoproteomics to enumerate the full set of dioxygenases and other proteins inhibited or covalently modified by D2-HG, succinate, and fumarate ([PMID: 41844803](#)), informing therapeutic strategy.

10. Key References

- Oncometabolites and TCA enzymes as cancer drivers — [PMID: 31085323](#); [PMID: 21301796](#); [PMID: 42327104](#); [PMID: 28104508](#)
- Primordial/reductive TCA and prebiotic chemistry — [PMID: 29420286](#); [PMID: 28970480](#); [PMID: 31712697](#)
- Succinyl-CoA synthetase mechanism and specificity — [PMID: 33645539](#); [PMID: 36189720](#); [PMID: 27478903](#); [PMID: 27487822](#); [PMID: 31315053](#)
- Aconitase / IRP1 moonlighting — [PMID: 8682202](#); [PMID: 10037708](#)
- Two fumarase classes — [PMID: 3282546](#); [PMID: 1329945](#); [PMID: 8473853](#); [PMID: 34083440](#)
- Complex II structure and non-canonical roles — [PMID: 37119852](#); [PMID: 33876763](#); [PMID: 41606097](#); [PMID: 34563205](#)
- Malate node alternatives — [PMID: 11092847](#); [PMID: 9660197](#)

- Pyruvate carboxylase anaplerosis — [PMID: 37603581](#); [PMID: 27732858](#); [PMID: 27890529](#); [PMID: 39725256](#)
 - IDH phospho-switch and glyoxylate bypass — [PMID: 22889914](#); [PMID: 30030382](#); [PMID: 23192354](#); [PMID: 29733852](#); [PMID: 33290880](#)
 - Cyanobacterial bypass and GABA shunt — [PMID: 22174252](#); [PMID: 24989231](#)
 - Citrate synthase catalytic strategy — [PMID: 7577912](#); [PMID: 8010958](#); [PMID: 23954305](#)
 - 2-oxoglutarate dehydrogenase complex — [PMID: 35530286](#); [PMID: 8720178](#)
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Prepared as a taxon-neutral review of the oxidative TCA cycle module. Claims are attributed to primary literature via PMID; cofactor, regulatory, and structural details are frequently organism-specific and should not be generalized across all lineages without direct evidence.

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